SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

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Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Developed a full-stack application using **Java, Spring Boot, MySQL, and React**, building and integrating **5+ REST APIs** for business workflows and data operations.
  - Designed relational database workflows and optimized **SQL queries, indexing, and data access patterns**, reducing backend response time by **25%** during testing.
  - Performed **data validation, query testing, debugging, and API integration**, ensuring reliable data flow between backend services, database layers, and application components.

- **All India Council for Technical Education (AICTE)** *July 2024 – September 2024*
  Google AI/ML Virtual Intern Remote
  - Developed and evaluated **8+ machine learning models** across supervised and unsupervised learning tasks using **5+ datasets**.
  - Applied **Python-based data preprocessing, cleaning, feature engineering, EDA, and hyperparameter tuning** to prepare datasets and improve model performance.
  - Evaluated real-world AI use cases using **model validation, performance analysis, and bias mitigation** techniques to improve reliability of data-driven solutions.

Projects

- **Cloud Data Engineering Pipeline & Lakehouse** *2026*
  Technologies: Python, PySpark, SQL, AWS S3, PostgreSQL, Pandas, Apache Spark
  - Engineered an end-to-end **batch ETL pipeline** for API and file-based ingestion, processing structured and semi-structured data using Python, Pandas, PySpark, and SQL.
  - Implemented **data cleansing, schema validation, deduplication, incremental processing, partitioning, and data quality checks** across ingestion and transformation stages.
  - Designed a **lakehouse-style architecture** using AWS S3 and PostgreSQL with optimized analytical tables for scalable downstream SQL workloads.

- **Enterprise Data Warehouse & Analytics Platform** *2026*
  Technologies: SQL, Python, PostgreSQL, Pandas, AWS, Streamlit
  - Designed a production-style **dimensional data warehouse** using star schema, fact and dimension tables, surrogate keys, and Slowly Changing Dimensions (SCD).
  - Developed complex SQL transformations using **CTEs, subqueries, joins, window functions, aggregations, CASE expressions, and reconciliation queries**.
  - Optimized analytical workloads using **indexing, query optimization, partition-aware processing, and performance-oriented data modeling**.

- **GenAI-Powered Data Engineering Assistant** *2026*
  Technologies: Python, SQL, PySpark, LangChain, LLMs, RAG, PostgreSQL
  - Built a GenAI assistant for **natural-language SQL generation, data profiling, pipeline documentation, and automated dataset summarization**.
  - Implemented an LLM-powered workflow combining **schema-aware retrieval, SQL generation, query validation, error handling, and structured responses**.
  - Integrated RAG with metadata and database context to improve **data-query relevance and contextual accuracy** across analytical datasets.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Developed and evaluated a deep learning pipeline using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Applied **data preprocessing, PCA, augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ accuracy** and **0.997 AUC**, improving performance by **7% over baseline methods** and MCC by **8%**.

Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

Technical Skills

- **Programming**: Python, SQL, Java, C++, JavaScript
- **Data Engineering**: PySpark, Apache Spark, ETL/ELT, Data Pipelines, Data Processing, Data Cleaning, Data Validation, Data Quality
- **SQL & Databases**: MySQL, PostgreSQL, MongoDB, SQL Joins, CTEs, Subqueries, Window Functions, Query Optimization, Indexing
- **Data Modeling & Warehousing**: Data Modeling, Data Warehousing, Star Schema, Snowflake Schema, Fact & Dimension Tables, OLTP, OLAP, SCD, Normalization
- **Cloud & Data Platforms**: AWS, Azure, GCP, Data Lakes, Cloud Storage, Cloud Data Services, Docker
- **Data Science & AI**: Pandas, NumPy, Scikit-learn, Machine Learning, EDA, Feature Engineering, LLMs, RAG, Prompt Engineering
- **Development & Tools**: Git, GitHub, Linux, Jupyter Notebook, REST APIs, FastAPI, Streamlit, VS Code